PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Community Engagement in research in Sub Saharan Africa: Approaches, barriers, facilitators, Ethical Considerations and the role of Gender - A systematic review protocol
AUTHORS	Akondeng, Claudine; Njamnshi, Wepnyu Y; Mandi, Henshaw Eyambe; Agbor, Valirie Ndip; Bain, Luchuo; Njamnshi, Alfred K.

VERSION 1 – REVIEW

REVIEWER	Anna Kalbarczyk Johns Hopkins University Bloomberg School of Public Health, International Health
REVIEW RETURNED	06-Nov-2021

GENERAL COMMENTS	Thank you for the opportunity to review this protocol. Below I have included a few comments I hope the authors find helpful: Abstract line 22 "suearch" should be "search" In the abstract it would be more helpful to indicate the "categories" or buckets of search terms, and then indicate the number and range of databases, rather than listing all of the databases. What is the rationale for restricting the search dates? Please include one. Why exclude commentaries? These could have valuable insights based on experiences with CE? I would just ask the authors to justify this choice and I also think this is a major limitation of the outlined approach. Similarly, 'human population health research' is quite vague. What is included in this field specifically? Neither gender nor ethics were included in the search strategy but these are part of two study objectives. What is the rationale? How were the databases selected? That is, why these and not others? The authors mention in the background that CE approaches, when used are often based on Western approaches. And increasingly systematic reviews are criticized in terms of their 'knowledge generation'. Particularly for community engagement I would encourage the authors to also consider other avenues to explore 'knowledge' for how best to do CE. And maybe take a close look at those articles that derive recommendations from the communities themselves, rather than the research team. This is part of why I

	argue that excluding commentaries is a critical limitation.

REVIEWER	Albino Kalolo Saint Francis University College of Health and Allied Sciences, Public Health
REVIEW RETURNED	08-Nov-2021

GENERAL COMMENTS	Reviewer comments This is a useful and clear description of a very important systematic review on community engagement in health research in Sub-Saharan Africa. There are some observations and suggestions for further strengthening the study 1) It is recommended that a through language check is done to address the typos and grammatic errors 2) Although the authors define community engagement in health research , they are silent on 1) which type of health research are they actually focusing? 2) Which steps/ strategies / activities indicate that there is community engagement in research 3) They point that in some cases Western models of community engagement can be used to engage communities in research . They are silent on 1) which (Western) models are these? 2) are there non-western models for community engagement in research ? , if yes which are they? 4) The sentence in page 4 , line 21-25, " In Sub-Saharan African settings, where the theory behind recommendations for community engagement is often not linked to empirical evidence, this study purposes ……" . is this the opinion of the authors or is backed by evidence? Given the fact that sub-Saharan Africa is not homogeneous , there are several countries with different culture, health systems and levels of development, the authors could be more careful and actually provide specific examples that cut across the different countries/ settings from east to west, north to south. 5) In the discussion section, there is a need to show if there are already such systematic studies in other settings (low and middle income as well as high income countries) and how the current study could provide a comparative picture of CE in health research in a stride forward to standardizing CE across settings 6) The statement in page 10 line 47-48: "The short-term impact of this project is that its results will serve as baseline data to inform stakeholders on how to engage communities in a gender-sensitive, ethical, culture-specific and sustainable way" could be more meaningful if the authors explain (preferably in the introduction/ methods section) what does it mean by 1) Engaging the community in a gender sensitive way 2) Engaging the community in an ethical way 3) engaging the community in a cultural specific way 4) Engaging the community in a sustainable way 7) Dissemination of research findings only to the Cameroonian government may be adequate given the fact that the study covers an entire sub-Saharan Africa region. The authors could rethink about their dissemination strategy

REVIEWER	David Snadden The University of British Columbia Faculty of Medicine, Family Practice
REVIEW RETURNED	12-Nov-2021

GENERAL COMMENTS	Thanks for the opportunity to review this work.

	Overall the protocol is well thought through and clearly explained. It is well written. Introduction. The introduction sets out some of the challenges inherent in community engaged research. The introduction frames the context for the study and explains the diversity of protocols and also raises the question of gender. One thing to consider when aiming for international publication is that gender is now being seen in the academic literature and in societal commentaries as a spectrum, while research focusing on male and female as binary concepts is beginning to use different terminology. While the context and culture of the research should determine your decisions, I offer one reference that may help you think about the use of gender/sex language in the protocol (1). I note that the methods of the review are focused on research carried out in Sub-Sharan Africa and while this is appropriate for the development of contextualized policy there may be a risk of limiting examples of other community engagement processes in other areas that may have transferrable elements of value to SSA. If the review remains focused in that area then I think there would need to be a more in-depth exploration of international literature in the introduction. My preference would be to widen the review itself to some extent as I will mention in the methods but that is a consideration for the study team and if the decision is made not to increase the scope of the review, then I would suggest more detail in the background. In this regard I offer two recent publications I was part of, one on a wide ranging community engagement process that collected research data, and another on a process for equitable engagement which may have some elements of interest as they use the WHO Towards Unity for Health model of engagement (2, 3). (https://bmjopen.bmj.com/content/11/10/e047165) The second of these references (Markham et al) has been accepted for publication by BMJOpen and should be published very soon. Objectives The objectives relate to the areas identified in the introduction. Research Questions The research questions also reflect the objectives. I have one observation and that is that if the focus is on what mechanisms have worked, or not worked, in SSA, in terms of community engagement will something be missing if the wider literature is excluded? Will the objectives constrain the imagination and creativity of the team as they try to find a way of developing a literature based framework adaptable to local contexts that will help others in SSA navigate the diversity of methods and protocols described in the introduction? Methods The systematic search databases seem comprehensive to me. The search strategy is confined to publications from SSA countries and for the years 2000-2001. This may not reveal many results in terms of articles and it might be worth having a contingency to broaden the time frame if only a few articles are identified. The search is also focused on published community engagement research using described models. Quoting from the eligibility criteria for article inclusion. “While several models of CE in research exist, such as participatory research, empowerment evaluation, participatory or

	community action research and participatory rapid appraisal,1 we will include any study that uses any of these models". In the introduction it states that there are a diversity of methods and I question whether focusing on actual CE methods used will allow the research team to explore the underpinning theoretical frameworks? One challenge I see in community engagement is the focus of research publications on standardized or best practice methods rather than on approaches that can be adapted to community contexts. I think the study will be more useful if engagement approaches are either discussed in the introduction, extracted from the research articles in the review (this might be most pragmatic approach, though may miss some) or added into the literature search strategy. I think the protocol will be improved if the team added processes that would allow the identification and inclusion of engagement approaches and frameworks that have been tried and have been found to be helpful and which are able to adapted and modified to local circumstances. Aside from the search strategy the methods for data extraction, assessment and synthesis are well written and clear, and in particular the three stage qualitative synthesis process will bring an important element of interpretation to the findings. Discussion This section is well written and has a very important focus on ethics, as well as engagement that encourages trust and mutual respect. It also touches on power dynamics. These are critical reflections and in many ways are also what are behind my comments in the methods on approaches, as it seems to me that it is the process of contextually adaptable engagement that is critical, not the rigidity of a published method and protocol, and this discussion section articulates this well. An example of an approach is one described by Boelen in the WHO publication Towards Unity for Health (4). This is an important study for the context of SSA and I would encourage a broader approach to the literature to include processes for engagement as well as methods and protocols, but in a way that would be contextually appropriate for SSA. A lot of thought and excellent work has gone into developing this protocol, and it may just be the first step in developing approaches that can be adapted to successfully work in SSA. It would not be unreasonable to carry out a scoping review first to understand the landscape of community engagement literature prior to a formal systematic review (5). 1. Heidari S, Babor TF, De Castro P, Tort S, Curno M. Sex and Gender Equity in Research: rationale for the SAGER guidelines and recommended use. Research Integrity and Peer Review. 2016;1(1):2. 2. Johnston CS, Belanger E, Wong K, Snadden D. How can rural community-engaged health services planning achieve sustainable healthcare system changes? BMJ Open. 2021;11(10):e047165. 3. Markham R, Hunt M, Wollard R, Oelke N, Snadden D, Strasser R, et al. Addressing Rural and Indigenous Health Inequities in Canada through Socially Accountable Health Partnerships. BMJOpen. 2021. 4. Boelen C. Towards Unity for Health: Challenges and Opportunities for Partnerships in Health Development. Geneva, Switzerland: World Health Organisation, WHO/EIP/OSD/2000.9; 2000. 5. Munn Z, Peters MDJ, Stern C, Tufanaru C, McArthur A,

	Aromataris E. Systematic review or scoping review? Guidance for authors when choosing between a systematic or scoping review approach. BMC Med Res Methodol. 2018;18(1):143.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1 Dr. Anna Kalbarczyk, Johns Hopkins University Bloomberg School of Public Health	
Reviewers comment	**Response**
Abstract line 22 "suearch" should be "search"	Corrected
In the abstract it would be more helpful to indicate the "categories" or buckets of search terms, and then indicate the number and range of 0020 databases, rather than listing all of the databases.	Accepted and corrected. The first sentence of the methods section of the abstract now reads: "A combination of key text words and medical subject headings such as "Community Engagement" or "Community Involvement" will be used to search 009 databases for all literature published between 1 January 2000 to 31 July 2021."
What is the rationale for restricting the search dates? Please include one.	Empirical research with focus on understanding engagement of communities in research is a recent phenomenon (2000 onwards). This is why we decided to limit our search from 2000 onwards. We think a 20 years search interval should be enough to provide relevant documented research findings to answer our review questions.
Why exclude commentaries? These could have valuable insights based on experiences with CE? I would just ask the authors to justify this choice and I also think this is a major limitation of the outlined approach.	Peer reviewed studies are used for the systematic review. Notwithstanding, we agree that valuable data of commentaries may be excluded from this review and this has been identified as a limitation of the study.
Similarly, 'human population health research' is quite vague. What is included in this field specifically?	Indeed, we intend here to talk about health-related research involving human subjects. The phrase has been modified to "community-based health research."
Neither gender nor ethics were included in the search strategy but these are part of two study objectives. What is the rationale?	The search strategy is focused on capturing CE and thus all studies on CE including gender and ethics will be extracted. The idea is to look at gender or ethics within the framework of CE.
How were the databases selected? That is, why these and not others?	We selected mostly databases that publish health-related research studies in sub-Saharan Africa. While there are multiple health-related

	databases, consider that the number of selected databases is comprehensive enough
The authors mention in the background that CE approaches, when used are often based on Western approaches. And increasingly systematic reviews are criticized in terms of their 'knowledge generation'. Particularly for community engagement I would encourage the authors to also consider other avenues to explore 'knowledge' for how best to do CE. And maybe take a close look at those articles that derive recommendations from the communities themselves, rather than the research team. This is part of why I argue that excluding commentaries is a critical limitation.	Noted. After reviewing effective interventions in SSA we will also do critical appraisals in our discussion with studies out of SSA. Subsequently, as follow up to this project and within the context of a different research study, we will conduct a focus group discussion with community engagement experts for qualitative data.
Reviewer: 2 Dr. Albino Kalolo, Saint Francis University College of Health and Allied Sciences	
Reviewers comment	**Response**
It is recommended that a through language check is done to address the typos and grammatical errors	Accepted and addressed
- Although the authors define community engagement in health research, they are silent on 1) Which type of health research are they actually focusing on? - Which steps/ strategies / activities indicate that there is community engagement in research	This review will focus on CE in community based participatory research. Strategies like community consultation, collaboration and communication are indicators of CE in research. Also, data from articles that discuss CE theories and frameworks will be extracted for analyses. The Data Item and Extraction section of the methodology summarizes how this will be done.
They point that in some cases Western models of community engagement can be used to engage communities in research. They are silent on 1) which (Western) models are these? 2) are there non-western models for community engagement in research, if yes which are they?	We have revised that statement. The point we are putting across here is, even though community engagement models are universal, their practical applications may be different across cultural contexts. The statement now reads, “Indeed, CE is mentioned in published research to describe community entry, consent and study participant retention. However, the actual CE activities are generally not well documented. Also, the fact that communities are heterogenous suggests that CE activities will not produce similar results across different contexts.”

	Our initial scoping of the literature revealed mostly western models, and these were generally cited in research carried out in SSA. These models shall be described in detail in the main systematic review, and most especially in the discussion section, to compare how they compare to our review findings with Sub - Saharan Africa focus.
The sentence in page 4 , line 21-25, " In Sub-Saharan African settings, where the theory behind recommendations for community engagement is often not linked to empirical evidence, this study purposes ……" . is this the opinion of the authors or is backed by evidence? Given the fact that sub-Saharan Africa is not homogeneous , there are several countries with different culture, health systems and levels of development, the authors could be more careful and actually provide specific examples that cut across the different countries / settings from east to west, north to south.	Accepted, we agree that SSA is not homogenous but during our analyses we will find similarities and differences across SSA countries. We have reviewed the paragraph which now reads, "Particularly, in Sub-Saharan African settings where there is a great community heterogeneity in terms of culture, this study proposes to synthesize available evidence across different countries in SSA on various practices for engaging communities in research and facilitators and challenges to this process."
In the discussion section, there is a need to show if there are already such systematic studies in other settings (low and middle income as well as high income countries) and how the current study could provide a comparative picture of CE in health research in a stride forward to standardizing CE across settings	We accept this suggestion and this will be done in detail in the final paper. For this protocol, the first sentence of the discussion has highlighted a systematic review done in high income settings with some of its limitations which are being addressed in this review. The statement reads "A similar review conducted documented many health systems interventions at the community level but didn't have a standard definition or reporting guidelines on describing community participation. Only a few studies in the review made references to CE definitions or frameworks"
The statement in page 10 line 47-48: "The short-term impact of this project is that its results will serve as baseline data to inform stakeholders on how to engage communities in a gender sensitive, ethical, culture-specific and sustainable way" could	Accepted. A section has been added in the introduction to define CE in the following modalities: gender sensitivity, ethics, culture and sustainability. (see pg 3, line 108-110; pg 4 line 119-121; 135-147)

be more meaningful if the authors explain (preferably in the introduction/ methods section) what does it mean by 1. Engaging the community in a gender sensitive way 2. Engaging the community in an ethical way 3. engaging the community in a cultural specific way 4. Engaging the community in a sustainable way	
Dissemination of research findings only to the Cameroonian government may be adequate given the fact that the study covers an entire sub-Saharan Africa region. The authors could rethink about their dissemination strategy	The results will be published in peer review journals and international scientific meetings accessible to all.
Reviewer: 3 Dr. David Snadden, The University of British Columbia Faculty of Medicine, The University of British Columbia Faculty of Medicine	
Reviewers comment	**Response**
Introduction. The introduction sets out some of the challenges inherent in community engaged research. The introduction frames the context for the study and explains the diversity of protocols and also raises the question of gender. One thing to consider when aiming for international publication is that gender is now being seen in the academic literature and in societal commentaries as a spectrum, while research focusing on male and female as binary concepts is beginning to use different terminology. While the context and culture of the research should determine your decisions, I offer one reference that may help you think about the use of gender/sex language in the protocol (1). I note that the methods of the review are focused on research carried out in Sub-Saharan Africa and while this is appropriate for the development of contextualized policy there may be a risk of limiting examples of other community engagement processes in other areas that may have transferrable elements of value to SSA. If the review remains focused in that area then I think there would need to be a more in-depth exploration of international literature in the introduction. My preference would be to widen the review itself to some extent as I will mention in the methods but that is a consideration for the study team and if the decision is made not to increase the scope of the review, then I would suggest more detail in the	Our review will extract data from already published studies which could present gender as binary concept or not. We will report the findings in each article and discuss its effects within the framework of community engagement. We agree with you that the context of the research will determine what data we get and guide the discussions. In the discussion section of our final article, we will discuss CE strategies from the data extracted with examples from other areas other than SSA.

background. In this regard I offer two recent publications I was part of, one on a wide ranging community engagement process that collected research data, and another on a process for equitable engagement which may have some elements of interest as they use the WHO Towards Unity for Health model of engagement (2, 3). (https://bmjopen.bmj.com/content/11/10/e047165) The second of these references (Markham et al) has been accepted for publication by BMJOpen and should be published very soon	
Research Questions The research questions also reflect the objectives. I have one observation and that is that if the focus is on what mechanisms have worked, or not worked, in SSA, in terms of community engagement will something be missing if the wider literature is excluded? Will the objectives constrain the imagination and creativity of the team as they try to find a way of developing a literature based framework adaptable to local contexts that will help others in SSA navigate the diversity of methods and protocols described in the introduction?	The focus will be community engagement in research in SSA but in the final paper, we will do a critical appraisal in our discussion with studies out of SSA.
Methods The systematic search databases seem comprehensive to me. The search strategy is confined to publications from SSA countries and for the years 2000-2021. This may not reveal many results in terms of articles and it might be worth having a contingency to broaden the time frame if only a few articles are identified. The search is also focused on published community engagement research using described models. Quoting from the eligibility criteria for article inclusion. “While several models of CE in research exist, such as participatory research, empowerment evaluation, participatory or community action research and participatory rapid appraisal,1 we will include any study that uses any of these models”. In the introduction it states that there are a diversity of methods and I question whether focusing on actual CE methods used will allow the research team to explore the underpinning theoretical frameworks? One challenge I see in community engagement is the focus of research publications on standardized or best practice methods rather than on approaches that can be adapted to community contexts. I think the study will be more useful if engagement approaches are either discussed in the introduction,	Accepted. In the introduction, a CE framework have been presented (see the last paragraph of page 4). Following data extraction, the results from the included articles will be presented and discussed in detail in the final manuscript. The discussion of the approaches and frameworks will be done in the discussion section of the final article.

extracted from the research articles in the review (this might be the most pragmatic approach, though may miss some) or added into the literature search strategy. I think the protocol will be improved if the team added processes that would allow the identification and inclusion of engagement approaches and frameworks that have been tried and have been found to be helpful and which are able to be adapted and modified to local circumstances.	
Discussion This section is well written and has a very important focus on ethics, as well as engagement that encourages trust and mutual respect. It also touches on power dynamics. These are critical reflections and in many ways are also what are behind my comments in the methods on approaches, as it seems to me that it is the process of contextually adaptable engagement that is critical, not the rigidity of a published method and protocol, and this discussion section articulates this well. An example of an approach is one described by Boelen in the WHO publication Towards Unity for Health (4). This is an important study for the context of SSA and I would encourage a broader approach to the literature to include processes for engagement as well as methods and protocols, but in a way that would be contextually appropriate for SSA. A lot of thought and excellent work has gone into developing this protocol, and it may just be the first step in developing approaches that can be adapted to successfully work in SSA. It would not be unreasonable to carry out a scoping review first to understand the landscape of community engagement literature prior to a formal systematic review (5).	We did a scoping search before building the protocol. However, we have included the suggestion on widening the background literature to enable us present CE concepts and approaches.

VERSION 2 – REVIEW

REVIEWER	Anna Kalbarczyk Johns Hopkins University Bloomberg School of Public Health, International Health
REVIEW RETURNED	22-Feb-2022

GENERAL COMMENTS	Thank you for responding to the original set of comments. While I appreciate seeing your rationale in the response to reviewers I think you should incorporate these rationales into the paper itself. For example, the rationale of date range restriction and selecting specific

	dates. I also think your search strategy may yield very many results - even if you do not include it here in the protocol, I recommend you consider adding a 'gender' concept to narrow your search.

REVIEWER	Albino Kalolo Saint Francis University College of Health and Allied Sciences, Public Health
REVIEW RETURNED	28-Feb-2022

GENERAL COMMENTS	I am satisfied with your response to the comments raised in the first review. I recommend the paper to be accepted. The following statement that appear both in the abstract and the methods section will have to be revised before the paper can be accepted for publication. " A combination of key text words and medical subject headings such as “Community Engagement” or “Community Involvement” will be used to search 009 databases" This sentence is somehow confusing because it is not clear of what you mean by key text words but also medical subject heading . With exception of the geographical search terms that you have clearly explained, please include the words that will be used as search terms for constructs related specifically to "community engagement in research".

REVIEWER	David Snadden The University of British Columbia Faculty of Medicine, Family Practice
REVIEW RETURNED	07-Feb-2022

GENERAL COMMENTS	All reviewer comments and suggestions have been addresses. Article reads well.

VERSION 2 – AUTHOR RESPONSE

Reviewer 3: Thank you

Reviewer 1: Accepted- Rationale in the response to reviewers will be incorporated into the paper itself. As concerns gender, this review will analyze gender considerations in all included papers, however, we did not want to limit our range of papers to review by including gender because very few papers clearly address gender considerations. We believe this will be so restrictive.

Reviewer 2: Thank you. All the keywords that will be used as search terms for constructs related specifically to "community engagement in research" are listed in table 1: "Community Participation/ or Community-Based Participatory Research/ or Community engagement.mp". We do not want to limit it to gender or ethics but get all papers that discuss community engagement.